

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Storage Limits			
Max Number of Drives (Total)	480	144	144
Internal Drives count	N/A	Min: 8 per Chassis; Max: 24 per Chassis;	Max: 24 per Chassis;
Max number of NVMe SSD Drives	Max: 240 per HA Pair;	Max: 48 per HA Pair;	Max: 72 per HA Pair;
Max number of SSD Drives	Max: 480 per HA Pair;	Max: 120 per HA Pair;	Max: 120 per HA Pair;
Max number of DS224C shelves	20 external	5 external	5 external
Max number of FS4483 shelves	N/A	N/A	N/A
Max number of NS224 shelves	10 external	1 internal + 1 external	1 internal + 2 external
Max number of NS224 w/ NSM100B shelves	10 external	1 external	2 external
Capacity Limits			
Marketing Maximum Raw Capacity (GB)			
Node	14,712,000	4,411,200	4,413,600
HA Pair	14,712,000	4,411,200	4,413,600
Core Cluster Limits			
Maximum number of storage virtual machines (SVMs) - NAS			
Cluster	1,024	512	512
Maximum number of LIFs - NAS			
Node	1,024	512	512
HA Pair	1,024	512	512
Cluster	4,096	4,096	4,096
Maximum number of connections - NAS/S3			
Node	100,000	100,000	100,000
HA Pair (Non-Scalable)	100,000	100,000	100,000
Maximum Number of VLANs			
Node	512	512	512
Maximum number of flexible volumes - NAS			
Node	2,500	1,000	1,000
HA Pair	5,000	2,000	2,000
Cluster	30,000	6,000	8,000
Maximum number of lock manager locked objects (CIFS and NFS combined)			
Node	3,000,000	1,200,000	1,200,000
HA Pair	3,000,000	1,200,000	1,200,000
Cluster	12,288,000	12,288,000	12,288,000
Maximum number of Intercluster LIFs			
Node	8	8	8
Minimum Root Aggregate (MiB)			
Node	191,488	191,488	191,488
Minimum Root Volume (GiB)			
Node	150	150	150
Maximum IPsec Policies			
SVM	256	N/A	N/A
Cluster	1,500	N/A	N/A
Maximum Number of IPspaces			
Cluster	512	512	512
Maximum IPsec Policies (If not using HW Offload)			
SVM	N/A	256	256
Cluster	N/A	1,500	1,500

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum IPsec Policies (If using HW Offload)			
SVM	N/A	40	40
Cluster	N/A	40	40
Maximum Number of Auditing-Enabled SVMs			
Cluster	400	400	400
Maximum Composite Aggregate Size (TiB)			
HA Pair	16,800	16,800	16,800
System Maximums and Limits			
Max Nodes per Cluster (NAS / SAN)	24 / 12	6 / 6	8 / 8
Max FabricPool Size	16800 TiB	16800 TiB	16800 TiB
Max FlexGroup Data Constituent Size	300 TiB	300 TiB	300 TiB
Max Infinite Volume Data Constituent Size	Not Supported	Not Supported	Not Supported
Min Root Volume Size	150 GiB	150 GiB	150 GiB
NetApp Volume Encryption	NVE	NVE	NVE
Processor			
Processor Model	64-bit 52-Core 1.7GHz	64-bit 8-core 2.1 GHz	64-bit 16-core 2.0 GHz
Processor Architecture	64 bit	64 bit	64 bit
Processor Speed	1.70 Ghz	2.10 Ghz	2.00 Ghz
Processor Count (Per Node)	2	1	1
Processor Count (Per Config)	4	2	2
Processor Cores (Per CPU)	52	8	16
Processor Cores (Per Node)	104	8	16
Processor Cores (Per Config)	208	16	32
Memory			
NVMEM (Per Node)	N/A	8 GB	8 GB
NVMEM (Per Config)	N/A	16 GB	16 GB
NVRAM (Per Node)	64 GB	N/A	N/A
NVRAM (Per Config)	128 GB	N/A	N/A
RAM (Per Node)	1024 GB	64 GB	64 GB
RAM (Per Config)	2048 GB	128 GB	128 GB
Onboard Ports			
Ethernet Ports	-	-	-
UTA2 Ports	-	-	-
Expansion Slots	9 x IO Module (per Chassis) 18 x IO Module (per System)	8 x IO Module (per System)	8 x IO Module (per System)
SAS Ports	-	-	-
Physical Characteristics			
Rack Units	4	2	2
Chassis Height	3.5" (8.89 cm)	3.43" (8.7 cm)	3.43" (8.7 cm)
Chassis Width with Mounting Flanges	19.02" (48.3 cm)	19.02" (48.3 cm)	19.02" (48.3 cm)
Chassis Width without Mounting Flanges	17.68" (44.9 cm)	17.6" (44.7 cm)	17.6" (44.7 cm)
Chassis Depth with Cable Mgmt	35.24" (89.5 cm)	-	-
Chassis Depth without Cable Mgmt	30.04" (76.3 cm)	22.05" (56 cm)	22.05" (56 cm)
Chassis Weight			
System Clearance Dimensions			
Front Clearance (Cooling)	6.03" (15.3 cm)	6.03" (15.3 cm)	6.03" (15.3 cm)
Front Clearance (Maintenance)	31.99" (81.2 cm)	24.03" (61 cm)	24.03" (61 cm)
Rear Clearance (Cooling)	6.03" (15.3 cm)	6.03" (15.3 cm)	6.03" (15.3 cm)
Rear Clearance (Maintenance)	11.03" (28 cm)	16.94" (43 cm)	16.94" (43 cm)

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Environmental Requirements			
Operating Temperature Range	50 to 95 deg F 10 to 35 deg C	50 to 95 deg F 10 to 35 deg C	50 to 95 deg F 10 to 35 deg C
Storage Temperature Range	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C
Transit Temperature Range	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C
Operating Relative Humidity	8 to 80 %	8 to 80 %	8 to 80 %
Storage Relative Humidity	10 to 95 %	10 to 95 %	10 to 95 %
Transit Relative Humidity	10 to 95 %	10 to 95 %	10 to 95 %
Operating Altitude Range	Up to 10000.0 ft Up to 3048.0 m	Up to 10000.0 ft Up to 3048.0 m	Up to 10000.0 ft Up to 3048.0 m
Storage Altitude Range	Up to 40000.0 ft Up to 12192.0 m	Up to 40000.0 ft Up to 12192.0 m	Up to 40000.0 ft Up to 12192.0 m
Transit Altitude Range	Up to 39989.8 ft Up to 12192 m	Up to 39989.8 ft Up to 12192 m	Up to 39989.8 ft Up to 12192 m
Acoustic Noise - Sound Power	8.5 Bels	8 Bels	8 Bels
Acoustic Noise - Sound Pressure	67.7 dBA	70.5 dBA	70.5 dBA
Input Power Voltage	40 to 48; 200 to 240;	40 to 48; 100 to 120; 200 to 240;	40 to 48; 200 to 240;
Storage OS Requirements (for selected major version)			
Recommended Version	9.19.1, 9.19.1P2	9.19.1, 9.19.1P2	9.19.1, 9.19.1P2
Minimum OS	9.19.1RC1	9.19.1RC1	9.19.1RC1
Maximum OS	9.19.1P2	9.19.1P2	9.19.1P2
Software and Firmware * indicates that firmware is bundled with Storage OS Version			
BIOS	20.8*	21.4.1, 21.3*	21.4.1, 21.3*
BMC	18.6*	19.4*	19.4*
Product Standards Compliance			
Certifications EMC/EMI	AMCA, FCC, ICES, KC, Morocco, VCCI	AMCA, FCC, ICES, KC, Morocco, VCCI	AMCA, FCC, ICES, KC, Morocco, VCCI
Certifications safety	BIS, CB, CSA, G_K_U-SoR, IRAM, NOM, NRCS, SONCAP, TBS	BIS, CB, CSA, G_K_U-SoR, IRAM, NOM, NRCS, SONCAP, TBS	BIS, CB, CSA, G_K_U-SoR, IRAM, NOM, NRCS, SONCAP, TBS
Certifications Safety/EMC/EMI	EAC, UKRSEPRO	EAC, UKRSEPRO	EAC, UKRSEPRO
Certifications Safety/EMC/EMI/RoHS	BSMI, CE DoC, UKCA DoC	BSMI, CE DoC, UKCA DoC	BSMI, CE DoC, UKCA DoC
Standards EMC/EMI	BS-EN-55032, BS-EN55035, CISPR 32, EN55022, EN55024, EN55032, EN55035, EN61000-3-2, EN61000-3-3, FCC Part 15 Class A, ICES-003, KS C 9832, KS C 9835	BS-EN-55024, BS-EN55035, CISPR 32, EN55022, EN55024, EN55032, EN55035, EN61000-3-2, EN61000-3-3, FCC Part 15 Class A, ICES-003, KS C 9832, KS C 9835	BS-EN-55032, BS-EN55035, CISPR 32, EN55022, EN55024, EN55032, EN55035, EN61000-3-2, EN61000-3-3, FCC Part 15 Class A, ICES-003, KS C 9832, KS C 9835
Standards Safety	ANSI/UL60950-1, ANSI/UL62368-1, BS-EN62368-1, CAN/CSA C22.2 No. 60950-1, CAN/CSA C22.2 No. 62368-1, CNS 15598-1, EN60825-1, EN62368-1, IEC 62368-1, IEC60950-1, IS 13252(part 1)	ANSI/UL60950-1, ANSI/UL62368-1, BS-EN62368-1, CAN/CSA C22.2 No. 60950-1, CAN/CSA C22.2 No. 62368-1, CNS 15598-1, EN60825-1, EN62368-1, IEC 62368-1, IEC60950-1, IS 13252(part 1)	ANSI/UL60950-1, ANSI/UL62368-1, BS-EN62368-1, CAN/CSA C22.2 No. 60950-1, CAN/CSA C22.2 No. 62368-1, CNS 15598-1, EN60825-1, EN62368-1, IEC 62368-1, IEC60950-1, IS 13252(part 1)
System Availability & Support			
Release Date	May 2024	Nov 2024	Nov 2024
End of Availability (EOA)	-	-	-
End of Support (EOS)	-	-	-
NFS Cluster Limits			
Maximum Number of NFS over TLS Connections			
Node	10,000	10,000	10,000
Maximum number of Export Policies			
Cluster	24,000	24,000	24,000

Specifications	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum number of Export Rules			
Cluster	200,000	200,000	200,000
Maximum NFSv4 Access Control Entries			
ACL (Access List)	1,024	1,024	1,024
Maximum number of client objects			
Node	100,000	100,000	100,000
HA Pair	100,000	100,000	100,000
Maximum Number of NFSv4 ByteLock Objects			
Node	26,624,000	1,000,000	1,000,000
Maximum Number of NFSv4 Client Objects			
Node	819,200	100,000	100,000
Maximum Number of NFSv4 Copy State Objects			
Node	8,192,000	10,000	10,000
Maximum Number of NFSv4 Deleg Objects			
Node	16,384,000	1,000,000	1,000,000
Maximum Number of NFSv4 Deleg State Objects			
Node	16,384,000	1,000,000	1,000,000
Maximum Number of NFSv4 Layout Objects			
Node	8,192,000	1,000,000	1,000,000
Maximum Number of NFSv4 Layout State Objects			
Node	8,192,000	1,000,000	1,000,000
Maximum Number of NFSv4 LayoutLock Objects			
Node	8,192,000	1,000,000	1,000,000
Maximum Number of NFSv4 Lock State Objects			
Node	16,384,000	4,000,000	4,000,000
Maximum Number of NFSv4 Open Objects			
Node	24,576,000	1,000,000	1,000,000
Maximum Number of NFSv4 Open State Objects			
Node	24,576,000	1,000,000	1,000,000
Maximum Number of NFSv4 Owner Objects			
Node	4,915,200	1,000,000	1,000,000
Maximum Number of NFSv4 Sessions			
Node	819,200	100,000	100,000
Maximum Number of NFSv4 String Objects			
Node	8,192,000	1,000,000	1,000,000
WAFL Cluster Limits			
Maximum Size of a 64-bit Aggregate (TiB)			
Aggregate	800	800	800
Maximum Size of a 64-bit Volume (TiB)			
Volume	300	300	300
Maximum file size in a 64-bit Volume (TiB)			
File	128	128	128
Maximum number of Volume Snapshot Copies			
Volume	1,023	1,023	1,023
Node	1,023,000	1,023,000	1,023,000
HA Pair	2,046,000	2,046,000	2,046,000
Maximum character length for Snapshot copy names			

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Snapshot	255	255	255
Maximum number of hard links			
File	100,000	100,000	100,000
Maximum number of inodes/files			
Volume	2,000,000,000	2,000,000,000	2,000,000,000
Maximum number of qtrees			
Volume	4,995	4,995	4,995
Node	100,000	100,000	100,000
HA Pair	200,000	200,000	200,000
Cluster	2,400,000	400,000	400,000
Maximum number of qtrees with Extended Performance Monitoring			
Cluster	50,000	50,000	50,000
Maximum number of concurrent DataMotion for Volumes (vol move) operations			
Node	16	16	16
HA Pair (Non-Scalable)	16	16	16
Maximum Depth of Clones			
Volume	1,000	999	999
Maximum Number of First Origin Clones			
Volume	1,023	999	999
Quality of Service Cluster Limits			
Maximum number of Policy Groups			
Cluster	12,000	12,000	12,000
Maximum number of QoS user workloads			
Cluster	40,000	40,000	40,000
SAN Cluster Limits			
Maximum number of storage virtual machines (SVMs) - SAN			
Cluster	1,000	750	1,000
Maximum number of flexible volumes - SAN			
Node	2,500	1,000	1,000
HA Pair	5,000	2,000	2,000
Cluster	30,000	6,000	8,000
Maximum number of LUNs			
Volume	1,024	1,024	1,024
Node	12,288	8,192	8,192
HA Pair	24,576	16,384	16,384
Cluster	98,304	49,152	65,536
Maximum number of LUN mappings			
igroup	4,096	4,096	4,096
Node	12,288	8,192	8,192
HA Pair	24,576	16,384	16,384
Cluster	98,304	49,152	65,536
Maximum LUN size (TiB)			
LUN	128	128	128
Maximum FC queue depth available			
Port	2,048	2,048	2,048
Maximum Number of SAN Hosts (ITNs)			
Port	512	512	512
Node	8,192	4,096	4,096
HA Pair	16,384	8,192	8,192
Cluster	65,536	24,576	32,768

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum number of LIFS - iSCSI			
Port	32	32	32
Portset	32	32	32
SVM	256	256	256
Node	1,024	1,024	1,024
HA Pair	2,048	2,048	2,048
Cluster	8,192	6,144	8,192
Maximum number of LIFS - FCP			
Port	32	32	32
Portset	32	32	32
SVM	256	256	256
Node	512	512	512
HA Pair	1,024	1,024	1,024
Cluster	4,096	3,072	4,096
Maximum number of igroups			
Node	4,096	2,048	2,048
HA Pair	8,192	4,096	4,096
Cluster	32,768	12,288	16,384
Maximum number of initiators			
igroup	256	256	256
Node	4,096	2,048	2,048
HA Pair	8,192	4,096	4,096
Cluster	32,768	12,288	16,384
Maximum number of portsets			
Node	4,096	2,048	2,048
HA Pair	8,192	4,096	4,096
Cluster	32,768	12,288	16,384
Maximum number of iSCSI sessions			
Node	8,192	4,096	4,096
HA Pair	16,384	8,192	8,192
Cluster	65,536	24,576	32,768
CIFS Cluster Limits			
Maximum number of connected shares			
Node	1,000,000	800,000	800,000
HA Pair	1,000,000	800,000	800,000
Maximum number of regular shares			
Cluster	300,000	300,000	300,000
Maximum number of open files			
Connection	65,536	65,536	65,536
Node	1,500,000	1,500,000	1,500,000
HA Pair	1,500,000	1,500,000	1,500,000
Maximum number of local users			
Cluster	35,000	35,000	35,000
Maximum number of local groups			
Cluster	320,000	320,000	320,000
Maximum number of local group members			
Cluster	640,000	640,000	640,000
Data Protection Cluster Limits			
Maximum number of NDMP sessions			
Node	36	36	36
HA Pair (Non-Scalable)	36	36	36
Maximum number of data protection (DP) mirrors and/or SnapVault relationships			

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Volume	1	1	1
Node	2,500	1,000	1,000
HA Pair	5,000	2,000	2,000
Cluster	15,000	6,000	8,000
Maximum number of load sharing (LS) mirrors			
NAS SVM	2	2	2
Maximum number of concurrent Snap Mirror or SnapVault transfers			
Node	250	100	100
Maximum fan-out from source for DP mirror			
Volume	20	8	8
Maximum fan-out from source for LS mirror			
NAS SVM	2	2	2
Maximum number of clusters that can be peered			
Cluster	400	400	400
Maximum Number of constituent volumes in a SnapMirror relationship			
FlexGroup	200	200	200
FlexGroup/Node	50	50	50
HA Pair	1,000	1,000	1,000
Cluster	6,000	6,000	6,000
Maximum number of SnapMirror Synchronous relationships			
Volume	1	1	1
Node	400	400	400
HA Pair	400	400	400
Maximum Number of Flexible Volumes in SVM DR			
SVM	300	300	300
HA Pair	1,000	1,000	1,000
Maximum Number of storage virtual machine disaster recovery (SVM DR) relationship			
Cluster	128	128	128
Maximum Number of Concurrent SVM Migration			
Cluster	1	1	1
Maximum Number of Volumes with SVM Migration			
SVM	400	400	400
Maximum Size of Cluster for SVM Migration			
Cluster	24	24	24
Maximum Number of FlexGroups and/or Consistency Groups supported in (SVM DR) relationship			
Cluster	32	32	32
Maximum Number of FlexGroups Constituents Volumes supported in (SVM DR) relationship			
FlexGroup	200	200	200
Maximum Number of Concurrent SnapDiff Sessions			
Node	64	64	64

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
HA Pair	128	128	128
NVMe/oF Cluster Limits			
Maximum number of storage virtual machines (SVMs) - NVMe/oF			
Cluster	32	32	32
Maximum number of flexible volumes - NVMe/oF			
Node	2,500	1,000	1,000
HA Pair	5,000	2,000	2,000
Cluster	30,000	6,000	8,000
Maximum number of namespaces			
Volume	1,024	1,024	1,024
Node	12,288	8,192	8,192
HA Pair	24,576	16,384	16,384
Cluster	98,304	49,152	65,536
Maximum number of subsystems			
SVM	8,192	8,192	8,192
Cluster	8,192	8,192	8,192
Maximum number of namespace mappings			
Subsystem	1,024	1,024	1,024
Node	12,288	8,192	8,192
HA Pair	24,576	16,384	16,384
Cluster	98,304	49,152	65,536
Maximum namespace size			
Namespace	128	128	128
Maximum I/O queue depth available - NVMe/FC			
Regular Priority Queue	32	32	32
High Priority Queue	64	32	32
Maximum I/O queue count available - NVMe/FC			
Regular Priority Controller	4	4	4
High Priority Controller	8	6	6
Maximum number of hosts			
Subsystem	64	64	64
Maximum number of LIFS - NVMe/oF			
Port	16	16	16
SVM	24	24	24
Node	32	32	32
HA Pair	32	32	32
Cluster	96	96	96
Node/SVM	2	2	2
Maximum number of controllers - NVMe/FC			
Port	256	256	256
Maximum I/O queue count available - NVMe/TCP			
Regular Priority Controller	2	2	2
High Priority Controller	8	4	4
Maximum I/O queue depth available - NVMe/TCP			
Regular Priority Queue	128	128	128
High Priority Queue	128	128	128
Maximum number of controllers - NVMe/TCP			
Node	2,048	128	128
FlexCache Cluster Limits			
Max caches from/of an origin volume (degree of fan-out)			
Origin	100	100	100
Max origin volumes			

Specifications	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Node	400	400	400
HA Pair	800	800	800
Max cache constituent volumes			
Node	800	800	800
HA Pair	1,600	1,600	1,600
Max cache volumes			
Node	400	400	400
HA Pair	800	800	800
Max origin constituent volumes			
Node	800	800	800
HA Pair	1,600	1,600	1,600
S3 Cluster Limits			
Maximum Number of Buckets			
FlexGroup Volume	1,000	1,000	1,000
Cluster	12,000	12,000	12,000
Maximum Number of Objects			
Bucket	400,000,000,000	400,000,000,000	400,000,000,000
Maximum Number of Users			
Object Store/StorageVM	4,000	4,000	4,000
Cluster	96,000	96,000	96,000
Maximum Size of the Bucket (PB)			
Vserver	60	60	60
Consistency Groups			
Maximum Number of CG Constituents Volumes supported in (SVM DR) relationship			
Consistency group	80	80	80
Maximum Number of Consistency Groups			
Cluster	15,000	15,000	15,000
Maximum Number of Consistency Groups and Parent Consistency Groups			
Cluster	15,000	15,000	15,000
Maximum Number of Parent Consistency Groups			
Cluster	15,000	15,000	15,000
Maximum Number of Child Consistency Groups in a Parent Consistency Group			
Parent consistency group	5	5	5
Maximum Number of Flexible Volumes in a Consistency Group			
Consistency group	80	80	80
Maximum Flexible Volumes across all the Child Consistency Groups under a Parent Consistency Group			
Parent consistency group	80	80	80
Maximum Number of Flexible Volumes in a Child Consistency Group			
Child consistency group	80	80	80
FlexGroup Rebalancing Limits			
Maximum Concurrent File Move at Destination Node			
Node	1,500	1,500	1,500

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum Number of Concurrent File Move Operations			
FlexGroup Constituent	1,500	1,500	1,500
SnapMirror Active Sync (asymmetric active/passive)			
Maximum number of Consistency Groups in SnapMirror Active Sync (asymmetric active/passive)relationships			
HA Pair (Non-Scalable)	50	50	50
Maximum Number of LUN Maps in SnapMirror SnapMirror Active Sync (asymmetric active/passive) relationships			
Node	2,048	2,048	2,048
Maximum Number of LUNs in SnapMirror Active Sync (asymmetric active/passive) relationships			
Volume	512	512	512
Maximum Number of Nodes in SnapMirror Active Sync (asymmetric active/passive) relationships			
Cluster	2	2	2
Maximum number of constituent volumes in SnapMirror Active Sync (asymmetric active/passive) relationships			
Consistency group	80	80	80
HA Pair (Non-Scalable)	400	400	400
Maximum Number of SAN LIFs in SnapMirror Active Sync (asymmetric active/passive) relationships			
Vserver	128	128	128
SnapMirror Active Sync (symmetric active/active)			
Maximum Number of Nodes in SnapMirror Active Sync relationships for NVMe			
Node	2	2	2
Maximum Number of NS Maps			
Consistency group	256	256	256
Maximum Number of Subsystem			
Consistency group	16	16	16
Maximum Number of Unique Namespaces Supported			
Node	4,096	4,096	4,096
Maximum Number of LUN Maps in SnapMirror Active Sync relationships			
Node	4,096	4,096	4,096

	AFF A1K Dual Chassis HA Pair	AFF A20 Single Chassis HA Pair	AFF A30 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum Number of Nodes in SnapMirror Active Sync relationships			
HA Pair	2	2	2
Cluster	4	4	4
Maximum Number of SAN LIFs in SnapMirror Active Sync relationships			
Vserver	128	128	128
Maximum Number of LUNs in SnapMirror Active Sync relationships			
Volume	512	512	512
Maximum number of Consistency Groups in SnapMirror Active Sync relationships			
HA Pair	50	50	50
HA Pair (Non-Scalable)	50	50	50
Maximum number of constituent volumes in SnapMirror Active Sync relationships			
Consistency group	80	80	80
HA Pair	400	400	400
HA Pair (Non-Scalable)	400	400	400
Foreign LUN Import			
Maximum Block Size			
Bytes	512	512	512
Maximum Number of Concurrent Foreign LUN Imports			
Node	8	8	8
Maximum Number of FLI Initiators			
Intercluster LIF	1	1	1
Maximum Number of Sessions			
Intercluster LIF	16	16	16
Node	32	32	32
Maximum Size of Imported LUN (TB)			
LUN	4	4	4

Specifications	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Storage Limits			
Max Number of Drives (Total)	144	240	240
Internal Drives count	Max: 24 per Chassis;	Max: 48 per Chassis;	Max: 48 per Chassis;
Max number of NVMe SSD Drives	Max: 120 per HA Pair;	Max: 240 per HA Pair;	Max: 240 per HA Pair;
Max number of SSD Drives	Max: 120 per HA Pair;	Max: 240 per HA Pair;	Max: 240 per HA Pair;
Max number of DS224C shelves	5 external	8 external	8 external
Max number of FS4483 shelves	N/A	1 internal	1 internal
Max number of NS224 shelves	1 internal + 4 external	8 external	8 external
Max number of NS224 w/ NSM100B shelves	4 external	8 external	8 external
Capacity Limits			
Marketing Maximum Raw Capacity (GB)			
Node	4,418,400	7,368,000	7,368,000
HA Pair	4,418,400	7,368,000	7,368,000
Core Cluster Limits			
Maximum number of storage virtual machines (SVMs) - NAS			
Cluster	512	1,024	1,024
Maximum number of LIFs - NAS			
Node	512	1,024	1,024
HA Pair	512	1,024	1,024
Cluster	4,096	4,096	4,096
Maximum number of connections - NAS/S3			
Node	100,000	100,000	100,000
HA Pair (Non-Scalable)	100,000	100,000	100,000
Maximum Number of VLANs			
Node	512	512	512
Maximum number of flexible volumes - NAS			
Node	2,500	2,500	2,500
HA Pair	5,000	5,000	5,000
Cluster	30,000	30,000	30,000
Maximum number of lock manager locked objects (CIFS and NFS combined)			
Node	1,200,000	3,000,000	3,000,000
HA Pair	1,200,000	3,000,000	3,000,000
Cluster	12,288,000	12,288,000	12,288,000
Maximum number of Intercluster LIFs			
Node	8	8	8
Minimum Root Aggregate (MiB)			
Node	191,488	191,488	191,488
Minimum Root Volume (GiB)			
Node	150	150	150
Maximum IPsec Policies			
SVM	256	256	256
Cluster	1,500	1,500	1,500
Maximum Number of IPspaces			
Cluster	512	512	512
Maximum IPsec Policies (If not using HW Offload)			
SVM	N/A	N/A	N/A
Cluster	N/A	N/A	N/A

	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum IPsec Policies (If using HW Offload)			
SVM	N/A	N/A	N/A
Cluster	N/A	N/A	N/A
Maximum Number of Auditing-Enabled SVMs			
Cluster	400	400	400
Maximum Composite Aggregate Size (TiB)			
HA Pair	16,800	16,800	16,800
System Maximums and Limits			
Max Nodes per Cluster (NAS / SAN)	24 / 12	24 / 12	24 / 12
Max FabricPool Size	16800 TiB	16800 TiB	16800 TiB
Max FlexGroup Data Constituent Size	300 TiB	300 TiB	300 TiB
Max Infinite Volume Data Constituent Size	Not Supported	Not Supported	Not Supported
Min Root Volume Size	150 GiB	150 GiB	150 GiB
NetApp Volume Encryption	NVE	NVE	NVE
Processor			
Processor Model	64-bit 24-core 1.9 GHz	64-bit 16-Core 2.0GHz	64-bit 32-Core 2.0GHz
Processor Architecture	64 bit	64 bit	64 bit
Processor Speed	1.90 Ghz	2.00 Ghz	2.00 Ghz
Processor Count (Per Node)	1	2	2
Processor Count (Per Config)	2	4	4
Processor Cores (Per CPU)	24	16	32
Processor Cores (Per Node)	24	32	64
Processor Cores (Per Config)	48	64	128
Memory			
NVMEM (Per Node)	16 GB	N/A	N/A
NVMEM (Per Config)	32 GB	N/A	N/A
NVRAM (Per Node)	N/A	32 GB	64 GB
NVRAM (Per Config)	N/A	64 GB	128 GB
RAM (Per Node)	128 GB	128 GB	1024 GB
RAM (Per Config)	256 GB	256 GB	2048 GB
Onboard Ports			
Ethernet Ports	-	-	-
UTA2 Ports	-	-	-
Expansion Slots	8 x IO Module (per System)	18 x IO Module (per System)	18 x IO Module (per System)
SAS Ports	-	-	-
Physical Characteristics			
Rack Units	2	4	4
Chassis Height	3.43" (8.7 cm)	6.89" (17.5 cm)	6.89" (17.5 cm)
Chassis Width with Mounting Flanges	19.02" (48.3 cm)	19.02" (48.3 cm)	19.02" (48.3 cm)
Chassis Width without Mounting Flanges	17.6" (44.7 cm)	17.6" (44.7 cm)	17.6" (44.7 cm)
Chassis Depth with Cable Mgmt	-	38.27" (97.2 cm)	38.27" (97.2 cm)
Chassis Depth without Cable Mgmt	22.05" (56 cm)	34.49" (87.6 cm)	34.49" (87.6 cm)
Chassis Weight			
System Clearance Dimensions			
Front Clearance (Cooling)	6.03" (15.3 cm)	6.03" (15.3 cm)	6.03" (15.3 cm)
Front Clearance (Maintenance)	24.03" (61 cm)	36.41" (92.4 cm)	36.41" (92.4 cm)
Rear Clearance (Cooling)	6.03" (15.3 cm)	6.03" (15.3 cm)	6.03" (15.3 cm)
Rear Clearance (Maintenance)	16.94" (43 cm)	36.01" (91.4 cm)	36.01" (91.4 cm)

	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Environmental Requirements			
Operating Temperature Range	50 to 95 deg F 10 to 35 deg C	50 to 95 deg F 10 to 35 deg C	50 to 95 deg F 10 to 35 deg C
Storage Temperature Range	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C
Transit Temperature Range	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C
Operating Relative Humidity	8 to 80 %	8 to 80 %	8 to 80 %
Storage Relative Humidity	10 to 95 %	10 to 95 %	10 to 95 %
Transit Relative Humidity	10 to 95 %	10 to 95 %	10 to 95 %
Operating Altitude Range	Up to 10000.0 ft Up to 3048.0 m	Up to 10000.0 ft Up to 3048.0 m	Up to 10000.0 ft Up to 3048.0 m
Storage Altitude Range	Up to 40000.0 ft Up to 12192.0 m	Up to 40000.0 ft Up to 12192.0 m	Up to 40000.0 ft Up to 12192.0 m
Transit Altitude Range	Up to 39989.8 ft Up to 12192 m	Up to 39989.8 ft Up to 12192 m	Up to 39989.8 ft Up to 12192 m
Acoustic Noise - Sound Power	8 Bels	8.6 Bels	8.6 Bels
Acoustic Noise - Sound Pressure	70.5 dBA	71.6 dBA	71.6 dBA
Input Power Voltage	40 to 48; 200 to 240;	40 to 48; 200 to 240;	40 to 48; 200 to 240;
Storage OS Requirements (for selected major version)			
Recommended Version	9.19.1, 9.19.1P2	9.19.1, 9.19.1P2	9.19.1, 9.19.1P2
Minimum OS	9.19.1RC1	9.19.1RC1	9.19.1RC1
Maximum OS	9.19.1P2	9.19.1P2	9.19.1P2
Software and Firmware * indicates that firmware is bundled with Storage OS Version			
BIOS	21.4.1, 21.3*	20.8*	20.8*
BMC	19.4*	18.6*	18.6*
Product Standards Compliance			
Certifications EMC/EMI	AMCA, FCC, ICES, KC, Morocco, VCCI	AMCA, FCC, ICES, KC, Morocco, VCCI	AMCA, FCC, ICES, KC, Morocco, VCCI
Certifications safety	BIS, CB, CSA, G_K_U-SoR, IRAM, NOM, NRCS, SONCAP, TBS	BIS, CB, CSA, G_K_U-SoR, IRAM, NOM, NRCS, SONCAP, TBS	BIS, CB, CSA, G_K_U-SoR, IRAM, NOM, NRCS, SONCAP, TBS
Certifications Safety/EMC/EMI	EAC, UKRSEPRO	EAC, UKRSEPRO	EAC, UKRSEPRO
Certifications Safety/EMC/EMI/RoHS	BSMI, CE DoC, UKCA DoC	BSMI, CE DoC, UKCA DoC	BSMI, CE DoC, UKCA DoC
Standards EMC/EMI	BS-EN-55032, BS-EN55035, CISPR 32, EN55022, EN55024, EN55032, EN55035, EN61000-3-2, EN61000-3-3, FCC Part 15 Class A, ICES-003, KS C 9832, KS C 9835	BS-EN-55032, BS-EN55035, CISPR 32, EN55022, EN55024, EN55032, EN55035, EN61000-3-2, EN61000-3-3, FCC Part 15 Class A, ICES-003, KS C 9832, KS C 9835	BS-EN-55032, BS-EN55035, CISPR 32, EN55022, EN55024, EN55032, EN55035, EN61000-3-2, EN61000-3-3, FCC Part 15 Class A, ICES-003, KS C 9832, KS C 9835
Standards Safety	ANSI/UL60950-1, ANSI/UL62368-1, BS-EN62368-1, CAN/CSA C22.2 No. 60950-1, CAN/CSA C22.2 No. 62368-1, CNS 15598-1, EN60825-1, EN62368-1, IEC 62368-1, IEC60950-1, IS 13252(part 1)	ANSI/UL60950-1, ANSI/UL62368-1, BS-EN62368-1, CAN/CSA C22.2 No. 60950-1, CAN/CSA C22.2 No. 62368-1, CNS 15598-1, EN60825-1, EN62368-1, IEC 62368-1, IEC60950-1, IS 13252(part 1)	ANSI/UL60950-1, ANSI/UL62368-1, BS-EN62368-1, CAN/CSA C22.2 No. 60950-1, CAN/CSA C22.2 No. 62368-1, CNS 15598-1, EN60825-1, EN62368-1, IEC 62368-1, IEC60950-1, IS 13252(part 1)
System Availability & Support			
Release Date	Nov 2024	May 2024	May 2024
End of Availability (EOA)	-	-	-
End of Support (EOS)	-	-	-
NFS Cluster Limits			
Maximum Number of NFS over TLS Connections			
Node	10,000	10,000	10,000
Maximum number of Export Policies			
Cluster	24,000	24,000	24,000

Specifications	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum number of Export Rules			
Cluster	200,000	200,000	200,000
Maximum NFSv4 Access Control Entries			
ACL (Access List)	1,024	1,024	1,024
Maximum number of client objects			
Node	100,000	100,000	100,000
HA Pair	100,000	100,000	100,000
Maximum Number of NFSv4 ByteLock Objects			
Node	1,000,000	1,000,000	26,624,000
Maximum Number of NFSv4 Client Objects			
Node	100,000	100,000	819,200
Maximum Number of NFSv4 Copy State Objects			
Node	10,000	10,000	8,192,000
Maximum Number of NFSv4 Deleg Objects			
Node	1,000,000	1,000,000	16,384,000
Maximum Number of NFSv4 Deleg State Objects			
Node	1,000,000	1,000,000	16,384,000
Maximum Number of NFSv4 Layout Objects			
Node	1,000,000	1,000,000	8,192,000
Maximum Number of NFSv4 Layout State Objects			
Node	1,000,000	1,000,000	8,192,000
Maximum Number of NFSv4 LayoutLock Objects			
Node	1,000,000	1,000,000	8,192,000
Maximum Number of NFSv4 Lock State Objects			
Node	1,000,000	1,000,000	16,384,000
Maximum Number of NFSv4 Open Objects			
Node	1,000,000	1,000,000	24,576,000
Maximum Number of NFSv4 Open State Objects			
Node	1,000,000	1,000,000	24,576,000
Maximum Number of NFSv4 Owner Objects			
Node	1,000,000	1,000,000	4,915,200
Maximum Number of NFSv4 Sessions			
Node	100,000	100,000	819,200
Maximum Number of NFSv4 String Objects			
Node	1,000,000	1,000,000	8,192,000
WAFL Cluster Limits			
Maximum Size of a 64-bit Aggregate (TiB)			
Aggregate	800	800	800
Maximum Size of a 64-bit Volume (TiB)			
Volume	300	300	300
Maximum file size in a 64-bit Volume (TiB)			
File	128	128	128
Maximum number of Volume Snapshot Copies			
Volume	1,023	1,023	1,023
Node	1,023,000	1,023,000	1,023,000
HA Pair	2,046,000	2,046,000	2,046,000
Maximum character length for Snapshot copy names			

	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Snapshot	255	255	255
Maximum number of hard links			
File	100,000	100,000	100,000
Maximum number of inodes/files			
Volume	2,000,000,000	2,000,000,000	2,000,000,000
Maximum number of qtrees			
Volume	4,995	4,995	4,995
Node	100,000	100,000	100,000
HA Pair	200,000	200,000	200,000
Cluster	400,000	2,400,000	2,400,000
Maximum number of qtrees with Extended Performance Monitoring			
Cluster	50,000	50,000	50,000
Maximum number of concurrent DataMotion for Volumes (vol move) operations			
Node	16	16	16
HA Pair (Non-Scalable)	16	16	16
Maximum Depth of Clones			
Volume	1,000	1,000	1,000
Maximum Number of First Origin Clones			
Volume	1,023	1,023	1,023
Quality of Service Cluster Limits			
Maximum number of Policy Groups			
Cluster	12,000	12,000	12,000
Maximum number of QoS user workloads			
Cluster	40,000	40,000	40,000
SAN Cluster Limits			
Maximum number of storage virtual machines (SVMs) - SAN			
Cluster	1,000	1,000	1,000
Maximum number of flexible volumes - SAN			
Node	2,500	2,500	2,500
HA Pair	5,000	5,000	5,000
Cluster	30,000	30,000	30,000
Maximum number of LUNs			
Volume	1,024	1,024	1,024
Node	12,288	12,288	12,288
HA Pair	24,576	24,576	24,576
Cluster	98,304	98,304	98,304
Maximum number of LUN mappings			
igroup	4,096	4,096	4,096
Node	12,288	12,288	12,288
HA Pair	24,576	24,576	24,576
Cluster	98,304	98,304	98,304
Maximum LUN size (TiB)			
LUN	128	128	128
Maximum FC queue depth available			
Port	2,048	2,048	2,048
Maximum Number of SAN Hosts (ITNs)			
Port	512	512	512
Node	8,192	8,192	8,192
HA Pair	16,384	16,384	16,384
Cluster	65,536	65,536	65,536

Specifications	AFF A50 Single Chassis HA Pair		AFF A70 Single Chassis HA Pair		AFF A90 Single Chassis HA Pair	
						
	ONTAP 9.19.1		ONTAP 9.19.1		ONTAP 9.19.1	
Maximum number of LIFS - iSCSI						
Port	32		32		32	
Portset	32		32		32	
SVM	256		256		256	
Node	1,024		1,024		1,024	
HA Pair	2,048		2,048		2,048	
Cluster	8,192		8,192		8,192	
Maximum number of LIFS - FCP						
Port	32		32		32	
Portset	32		32		32	
SVM	256		256		256	
Node	512		512		512	
HA Pair	1,024		1,024		1,024	
Cluster	4,096		4,096		4,096	
Maximum number of igroups						
Node	4,096		4,096		4,096	
HA Pair	8,192		8,192		8,192	
Cluster	32,768		32,768		32,768	
Maximum number of initiators						
igroup	256		256		256	
Node	4,096		4,096		4,096	
HA Pair	8,192		8,192		8,192	
Cluster	32,768		32,768		32,768	
Maximum number of portsets						
Node	4,096		4,096		4,096	
HA Pair	8,192		8,192		8,192	
Cluster	32,768		32,768		32,768	
Maximum number of iSCSI sessions						
Node	8,192		8,192		8,192	
HA Pair	16,384		16,384		16,384	
Cluster	65,536		65,536		65,536	
CIFS Cluster Limits						
Maximum number of connected shares						
Node	800,000		1,000,000		1,000,000	
HA Pair	800,000		1,000,000		1,000,000	
Maximum number of regular shares						
Cluster	300,000		300,000		300,000	
Maximum number of open files						
Connection	65,536		65,536		65,536	
Node	1,500,000		1,500,000		1,500,000	
HA Pair	1,500,000		1,500,000		1,500,000	
Maximum number of local users						
Cluster	35,000		35,000		35,000	
Maximum number of local groups						
Cluster	320,000		320,000		320,000	
Maximum number of local group members						
Cluster	640,000		640,000		640,000	
Data Protection Cluster Limits						
Maximum number of NDMP sessions						
Node	36		36		36	
HA Pair (Non-Scalable)	36		36		36	
Maximum number of data protection (DP) mirrors and/or SnapVault relationships						

	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Volume	1	1	1
Node	1,000	2,500	2,500
HA Pair	2,000	5,000	5,000
Cluster	15,000	15,000	15,000
Maximum number of load sharing (LS) mirrors			
NAS SVM	2	2	2
Maximum number of concurrent Snap Mirror or SnapVault transfers			
Node	100	250	250
Maximum fan-out from source for DP mirror			
Volume	8	20	20
Maximum fan-out from source for LS mirror			
NAS SVM	2	2	2
Maximum number of clusters that can be peered			
Cluster	400	400	400
Maximum Number of constituent volumes in a SnapMirror relationship			
FlexGroup	200	200	200
FlexGroup/Node	50	50	50
HA Pair	1,000	1,000	1,000
Cluster	6,000	6,000	6,000
Maximum number of SnapMirror Synchronous relationships			
Volume	1	1	1
Node	400	400	400
HA Pair	400	400	400
Maximum Number of Flexible Volumes in SVM DR			
SVM	300	300	300
HA Pair	1,000	1,000	1,000
Maximum Number of storage virtual machine disaster recovery (SVM DR) relationship			
Cluster	128	128	128
Maximum Number of Concurrent SVM Migration			
Cluster	1	1	1
Maximum Number of Volumes with SVM Migration			
SVM	400	400	400
Maximum Size of Cluster for SVM Migration			
Cluster	24	24	24
Maximum Number of FlexGroups and/or Consistency Groups supported in (SVM DR) relationship			
Cluster	32	32	32
Maximum Number of FlexGroups Constituents Volumes supported in (SVM DR) relationship			
FlexGroup	200	200	200
Maximum Number of Concurrent SnapDiff Sessions			
Node	64	64	64

	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
HA Pair	128	128	128
NVMe/oF Cluster Limits			
Maximum number of storage virtual machines (SVMs) - NVMe/oF			
Cluster	32	32	32
Maximum number of flexible volumes - NVMe/oF			
Node	2,500	2,500	2,500
HA Pair	5,000	5,000	5,000
Cluster	30,000	30,000	30,000
Maximum number of namespaces			
Volume	1,024	1,024	1,024
Node	12,288	12,288	12,288
HA Pair	24,576	24,576	24,576
Cluster	98,304	98,304	98,304
Maximum number of subsystems			
SVM	8,192	8,192	8,192
Cluster	8,192	8,192	8,192
Maximum number of namespace mappings			
Subsystem	1,024	1,024	1,024
Node	12,288	12,288	12,288
HA Pair	24,576	24,576	24,576
Cluster	98,304	98,304	98,304
Maximum namespace size			
Namespace	128	128	128
Maximum I/O queue depth available - NVMe/FC			
Regular Priority Queue	32	32	32
High Priority Queue	32	32	64
Maximum I/O queue count available - NVMe/FC			
Regular Priority Controller	4	4	4
High Priority Controller	6	6	8
Maximum number of hosts			
Subsystem	64	64	64
Maximum number of LIFS - NVMe/oF			
Port	16	16	16
SVM	24	24	24
Node	32	32	32
HA Pair	32	32	32
Cluster	96	96	96
Node/SVM	2	2	2
Maximum number of controllers - NVMe/FC			
Port	256	256	256
Maximum I/O queue count available - NVMe/TCP			
Regular Priority Controller	2	2	2
High Priority Controller	4	4	8
Maximum I/O queue depth available - NVMe/TCP			
Regular Priority Queue	128	128	128
High Priority Queue	128	128	128
Maximum number of controllers - NVMe/TCP			
Node	256	512	2,048
FlexCache Cluster Limits			
Max caches from/of an origin volume (degree of fan-out)			
Origin	100	100	100
Max origin volumes			

Specifications	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Node	400	400	400
HA Pair	800	800	800
Max cache constituent volumes			
Node	800	800	800
HA Pair	1,600	1,600	1,600
Max cache volumes			
Node	400	400	400
HA Pair	800	800	800
Max origin constituent volumes			
Node	800	800	800
HA Pair	1,600	1,600	1,600
S3 Cluster Limits			
Maximum Number of Buckets			
FlexGroup Volume	1,000	1,000	1,000
Cluster	12,000	12,000	12,000
Maximum Number of Objects			
Bucket	400,000,000,000	400,000,000,000	400,000,000,000
Maximum Number of Users			
Object Store/StorageVM	4,000	4,000	4,000
Cluster	96,000	96,000	96,000
Maximum Size of the Bucket (PB)			
Vserver	60	60	60
Consistency Groups			
Maximum Number of CG Constituents Volumes supported in (SVM DR) relationship			
Consistency group	80	80	80
Maximum Number of Consistency Groups			
Cluster	15,000	15,000	15,000
Maximum Number of Consistency Groups and Parent Consistency Groups			
Cluster	15,000	15,000	15,000
Maximum Number of Parent Consistency Groups			
Cluster	15,000	15,000	15,000
Maximum Number of Child Consistency Groups in a Parent Consistency Group			
Parent consistency group	5	5	5
Maximum Number of Flexible Volumes in a Consistency Group			
Consistency group	80	80	80
Maximum Flexible Volumes across all the Child Consistency Groups under a Parent Consistency Group			
Parent consistency group	80	80	80
Maximum Number of Flexible Volumes in a Child Consistency Group			
Child consistency group	80	80	80
FlexGroup Rebalancing Limits			
Maximum Concurrent File Move at Destination Node			
Node	1,500	1,500	1,500

	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum Number of Concurrent File Move Operations			
FlexGroup Constituent	1,500	1,500	1,500
SnapMirror Active Sync (asymmetric active/passive)			
Maximum number of Consistency Groups in SnapMirror Active Sync (asymmetric active/passive)relationships			
HA Pair (Non-Scalable)	50	50	50
Maximum Number of LUN Maps in SnapMirror SnapMirror Active Sync (asymmetric active/passive) relationships			
Node	2,048	2,048	2,048
Maximum Number of LUNs in SnapMirror Active Sync (asymmetric active/passive) relationships			
Volume	512	512	512
Maximum Number of Nodes in SnapMirror Active Sync (asymmetric active/passive) relationships			
Cluster	2	2	2
Maximum number of constituent volumes in SnapMirror Active Sync (asymmetric active/passive) relationships			
Consistency group	80	80	80
HA Pair (Non-Scalable)	400	400	400
Maximum Number of SAN LIFs in SnapMirror Active Sync (asymmetric active/passive) relationships			
Vserver	128	128	128
SnapMirror Active Sync (symmetric active/active)			
Maximum Number of Nodes in SnapMirror Active Sync relationships for NVMe			
Node	2	2	2
Maximum Number of NS Maps			
Consistency group	256	256	256
Maximum Number of Subsystem			
Consistency group	16	16	16
Maximum Number of Unique Namespaces Supported			
Node	4,096	4,096	4,096
Maximum Number of LUN Maps in SnapMirror Active Sync relationships			
Node	4,096	4,096	4,096

	AFF A50 Single Chassis HA Pair	AFF A70 Single Chassis HA Pair	AFF A90 Single Chassis HA Pair
Specifications			
	ONTAP 9.19.1	ONTAP 9.19.1	ONTAP 9.19.1
Maximum Number of Nodes in SnapMirror Active Sync relationships			
HA Pair	2	2	2
Cluster	4	4	4
Maximum Number of SAN LIFs in SnapMirror Active Sync relationships			
Vserver	128	128	128
Maximum Number of LUNs in SnapMirror Active Sync relationships			
Volume	512	512	512
Maximum number of Consistency Groups in SnapMirror Active Sync relationships			
HA Pair	50	50	50
HA Pair (Non-Scalable)	50	50	50
Maximum number of constituent volumes in SnapMirror Active Sync relationships			
Consistency group	80	80	80
HA Pair	400	400	400
HA Pair (Non-Scalable)	400	400	400
Foreign LUN Import			
Maximum Block Size			
Bytes	512	512	512
Maximum Number of Concurrent Foreign LUN Imports			
Node	8	8	8
Maximum Number of FLI Initiators			
Intercluster LIF	1	1	1
Maximum Number of Sessions			
Intercluster LIF	16	16	16
Node	32	32	32
Maximum Size of Imported LUN (TB)			
LUN	4	4	4

Specifications
Storage Limits
Max Number of Drives (Total)
Internal Drives count
Max number of NVMe SSD Drives
Max number of SSD Drives
Max number of DS224C shelves
Max number of FS4483 shelves
Max number of NS224 shelves
Max number of NS224 w/ NSM100B shelves
Capacity Limits
Marketing Maximum Raw Capacity (GB)
Node
HA Pair
Core Cluster Limits
Maximum number of storage virtual machines (SVMs) - NAS
Cluster
Maximum number of LIFs - NAS
Node
HA Pair
Cluster
Maximum number of connections - NAS/S3
Node
HA Pair (Non-Scalable)
Maximum Number of VLANs
Node
Maximum number of flexible volumes - NAS
Node
HA Pair
Cluster
Maximum number of lock manager locked objects (CIFS and NFS combined)
Node
HA Pair
Cluster
Maximum number of Intercluster LIFs
Node
Minimum Root Aggregate (MiB)
Node
Minimum Root Volume (GiB)
Node
Maximum IPsec Policies
SVM
Cluster
Maximum Number of IPspaces
Cluster
Maximum IPsec Policies (If not using HW Offload)
SVM
Cluster

Specifications
Maximum IPsec Policies (If using HW Offload)
SVM
Cluster
Maximum Number of Auditing-Enabled SVMs
Cluster
Maximum Composite Aggregate Size (TiB)
HA Pair
System Maximums and Limits
Max Nodes per Cluster (NAS / SAN)
Max FabricPool Size
Max FlexGroup Data Constituent Size
Max Infinite Volume Data Constituent Size
Min Root Volume Size
NetApp Volume Encryption
Processor
Processor Model
Processor Architecture
Processor Speed
Processor Count (Per Node)
Processor Count (Per Config)
Processor Cores (Per CPU)
Processor Cores (Per Node)
Processor Cores (Per Config)
Memory
NVMEM (Per Node)
NVMEM (Per Config)
NVRAM (Per Node)
NVRAM (Per Config)
RAM (Per Node)
RAM (Per Config)
Onboard Ports
Ethernet Ports
UTA2 Ports
Expansion Slots
SAS Ports
Physical Characteristics
Rack Units
Chassis Height
Chassis Width with Mounting Flanges
Chassis Width without Mounting Flanges
Chassis Depth with Cable Mgmt
Chassis Depth without Cable Mgmt
Chassis Weight
System Clearance Dimensions
Front Clearance (Cooling)
Front Clearance (Maintenance)
Rear Clearance (Cooling)
Rear Clearance (Maintenance)

Specifications
Environmental Requirements
Operating Temperature Range
Storage Temperature Range
Transit Temperature Range
Operating Relative Humidity
Storage Relative Humidity
Transit Relative Humidity
Operating Altitude Range
Storage Altitude Range
Transit Altitude Range
Acoustic Noise - Sound Power
Acoustic Noise - Sound Pressure
Input Power Voltage
Storage OS Requirements (for selected major version)
Recommended Version
Minimum OS
Maximum OS
Software and Firmware * indicates that firmware is bundled with Storage OS Version
BIOS
BMC
Product Standards Compliance
Certifications EMC/EMI
Certifications safety
Certifications Safety/EMC/EMI
Certifications Safety/EMC/EMI/RoHS
Standards EMC/EMI
Standards Safety
System Availability & Support
Release Date
End of Availability (EOA)
End of Support (EOS)
NFS Cluster Limits
Maximum Number of NFS over TLS Connections
Node
Maximum number of Export Policies
Cluster

Specifications
Maximum number of Export Rules
Cluster
Maximum NFSv4 Access Control Entries
ACL (Access List)
Maximum number of client objects
Node
HA Pair
Maximum Number of NFSv4 ByteLock Objects
Node
Maximum Number of NFSv4 Client Objects
Node
Maximum Number of NFSv4 Copy State Objects
Node
Maximum Number of NFSv4 Deleg Objects
Node
Maximum Number of NFSv4 Deleg State Objects
Node
Maximum Number of NFSv4 Layout Objects
Node
Maximum Number of NFSv4 Layout State Objects
Node
Maximum Number of NFSv4 LayoutLock Objects
Node
Maximum Number of NFSv4 Lock State Objects
Node
Maximum Number of NFSv4 Open Objects
Node
Maximum Number of NFSv4 Open State Objects
Node
Maximum Number of NFSv4 Owner Objects
Node
Maximum Number of NFSv4 Sessions
Node
Maximum Number of NFSv4 String Objects
Node
WAFL Cluster Limits
Maximum Size of a 64-bit Aggregate (TiB)
Aggregate
Maximum Size of a 64-bit Volume (TiB)
Volume
Maximum file size in a 64-bit Volume (TiB)
File
Maximum number of Volume Snapshot Copies
Volume
Node
HA Pair
Maximum character length for Snapshot copy names

Specifications						
Snapshot						
Maximum number of hard links						
File						
Maximum number of inodes/files						
Volume						
Maximum number of qtrees						
Volume						
Node						
HA Pair						
Cluster						
Maximum number of qtrees with Extended Performance Monitoring						
Cluster						
Maximum number of concurrent DataMotion for Volumes (vol move) operations						
Node						
HA Pair (Non-Scalable)						
Maximum Depth of Clones						
Volume						
Maximum Number of First Origin Clones						
Volume						
Quality of Service Cluster Limits						
Maximum number of Policy Groups						
Cluster						
Maximum number of QoS user workloads						
Cluster						
SAN Cluster Limits						
Maximum number of storage virtual machines (SVMs) - SAN						
Cluster						
Maximum number of flexible volumes - SAN						
Node						
HA Pair						
Cluster						
Maximum number of LUNs						
Volume						
Node						
HA Pair						
Cluster						
Maximum number of LUN mappings						
igroup						
Node						
HA Pair						
Cluster						
Maximum LUN size (TiB)						
LUN						
Maximum FC queue depth available						
Port						
Maximum Number of SAN Hosts (ITNs)						
Port						
Node						
HA Pair						
Cluster						

Specifications
Maximum number of LIFS - iSCSI
Port
Portset
SVM
Node
HA Pair
Cluster
Maximum number of LIFS - FCP
Port
Portset
SVM
Node
HA Pair
Cluster
Maximum number of igroups
Node
HA Pair
Cluster
Maximum number of initiators
igroup
Node
HA Pair
Cluster
Maximum number of portsets
Node
HA Pair
Cluster
Maximum number of iSCSI sessions
Node
HA Pair
Cluster
CIFS Cluster Limits
Maximum number of connected shares
Node
HA Pair
Maximum number of regular shares
Cluster
Maximum number of open files
Connection
Node
HA Pair
Maximum number of local users
Cluster
Maximum number of local groups
Cluster
Maximum number of local group members
Cluster
Data Protection Cluster Limits
Maximum number of NDMP sessions
Node
HA Pair (Non-Scalable)
Maximum number of data protection (DP) mirrors and/or SnapVault relationships

Specifications
Volume
Node
HA Pair
Cluster
Maximum number of load sharing (LS) mirrors
NAS SVM
Maximum number of concurrent Snap Mirror or SnapVault transfers
Node
Maximum fan-out from source for DP mirror
Volume
Maximum fan-out from source for LS mirror
NAS SVM
Maximum number of clusters that can be peered
Cluster
Maximum Number of constituent volumes in a SnapMirror relationship
FlexGroup
FlexGroup/Node
HA Pair
Cluster
Maximum number of SnapMirror Synchronous relationships
Volume
Node
HA Pair
Maximum Number of Flexible Volumes in SVM DR
SVM
HA Pair
Maximum Number of storage virtual machine disaster recovery (SVM DR) relationship
Cluster
Maximum Number of Concurrent SVM Migration
Cluster
Maximum Number of Volumes with SVM Migration
SVM
Maximum Size of Cluster for SVM Migration
Cluster
Maximum Number of FlexGroups and/or Consistency Groups supported in (SVM DR) relationship
Cluster
Maximum Number of FlexGroups Constituents Volumes supported in (SVM DR) relationship
FlexGroup
Maximum Number of Concurrent SnapDiff Sessions
Node

Specifications
HA Pair
NVMe/oF Cluster Limits
Maximum number of storage virtual machines (SVMs) - NVMe/oF
Cluster
Maximum number of flexible volumes - NVMe/oF
Node
HA Pair
Cluster
Maximum number of namespaces
Volume
Node
HA Pair
Cluster
Maximum number of subsystems
SVM
Cluster
Maximum number of namespace mappings
Subsystem
Node
HA Pair
Cluster
Maximum namespace size
Namespace
Maximum I/O queue depth available - NVMe/FC
Regular Priority Queue
High Priority Queue
Maximum I/O queue count available - NVMe/FC
Regular Priority Controller
High Priority Controller
Maximum number of hosts
Subsystem
Maximum number of LIFS - NVMe/oF
Port
SVM
Node
HA Pair
Cluster
Node/SVM
Maximum number of controllers - NVMe/FC
Port
Maximum I/O queue count available - NVMe/TCP
Regular Priority Controller
High Priority Controller
Maximum I/O queue depth available - NVMe/TCP
Regular Priority Queue
High Priority Queue
Maximum number of controllers - NVMe/TCP
Node
FlexCache Cluster Limits
Max caches from/of an origin volume (degree of fan-out)
Origin
Max origin volumes

Specifications
Node
HA Pair
Max cache constituent volumes
Node
HA Pair
Max cache volumes
Node
HA Pair
Max origin constituent volumes
Node
HA Pair
S3 Cluster Limits
Maximum Number of Buckets
FlexGroup Volume Cluster
Maximum Number of Objects
Bucket
Maximum Number of Users
Object Store/StorageVM Cluster
Maximum Size of the Bucket (PB)
Vserver
Consistency Groups
Maximum Number of CG Constituents Volumes supported in (SVM DR) relationship
Consistency group
Maximum Number of Consistency Groups
Cluster
Maximum Number of Consistency Groups and Parent Consistency Groups
Cluster
Maximum Number of Parent Consistency Groups
Cluster
Maximum Number of Child Consistency Groups in a Parent Consistency Group
Parent consistency group
Maximum Number of Flexible Volumes in a Consistency Group
Consistency group
Maximum Flexible Volumes across all the Child Consistency Groups under a Parent Consistency Group
Parent consistency group
Maximum Number of Flexible Volumes in a Child Consistency Group
Child consistency group
FlexGroup Rebalancing Limits
Maximum Concurrent File Move at Destination Node
Node

Specifications
Maximum Number of Concurrent File Move Operations
FlexGroup Constituent
SnapMirror Active Sync (asymmetric active/passive)
Maximum number of Consistency Groups in SnapMirror Active Sync (asymmetric active/passive)relationships
HA Pair (Non-Scalable)
Maximum Number of LUN Maps in SnapMirror SnapMirror Active Sync (asymmetric active/passive) relationships
Node
Maximum Number of LUNs in SnapMirror Active Sync (asymmetric active/passive) relationships
Volume
Maximum Number of Nodes in SnapMirror Active Sync (asymmetric active/passive) relationships
Cluster
Maximum number of constituent volumes in SnapMirror Active Sync (asymmetric active/passive) relationships
Consistency group
HA Pair (Non-Scalable)
Maximum Number of SAN LIFs in SnapMirror Active Sync (asymmetric active/passive) relationships
Vserver
SnapMirror Active Sync (symmetric active/active)
Maximum Number of Nodes in SnapMirror Active Sync relationships for NVMe
Node
Maximum Number of NS Maps
Consistency group
Maximum Number of Subsystem
Consistency group
Maximum Number of Unique Namespaces Supported
Node
Maximum Number of LUN Maps in SnapMirror Active Sync relationships
Node

Specifications						
Maximum Number of Nodes in SnapMirror Active Sync relationships						
HA Pair						
Cluster						
Maximum Number of SAN LIFs in SnapMirror Active Sync relationships						
Vserver						
Maximum Number of LUNs in SnapMirror Active Sync relationships						
Volume						
Maximum number of Consistency Groups in SnapMirror Active Sync relationships						
HA Pair						
HA Pair (Non-Scalable)						
Maximum number of constituent volumes in SnapMirror Active Sync relationships						
Consistency group						
HA Pair						
HA Pair (Non-Scalable)						
Foreign LUN Import						
Maximum Block Size						
Bytes						
Maximum Number of Concurrent Foreign LUN Imports						
Node						
Maximum Number of FLI Initiators						
Intercluster LIF						
Maximum Number of Sessions						
Intercluster LIF						
Node						
Maximum Size of Imported LUN (TB)						
LUN						